

Mathematical Analysis 2

Metric spaces 1

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1 Examples of metric spaces

The following definitions give metric spaces (M, d) . Verify this!

(a) **Absolute value.**

$$M = \mathbb{R}, \quad d(x, y) = |x - y|.$$

For parts (b)–(e), let $M = \mathbb{R}^n$, where $n \in \mathbb{N}$ (with $n = 2$ in part (e)).

(b) **Euclidean metric.**

$$d_2(x, y) = \sqrt{\sum_{i=1}^n |x_i - y_i|^2}.$$

The Cauchy–Bunyakovsky–Schwarz inequality is

$$\sum_{i=1}^n a_i b_i \leq \sqrt{\sum_{i=1}^n a_i^2} \sqrt{\sum_{i=1}^n b_i^2}.$$

Generalization:

$$d_p(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}, \quad p \in \mathbb{R}, \quad p \geq 1.$$

(c) **Manhattan (sum) metric.**

$$d_1(x, y) = \sum_{i=1}^n |x_i - y_i|.$$

(d) **Maximum metric.**

$$d_{\max}(x, y) = \max\{|x_i - y_i| : 1 \leq i \leq n\}.$$

(e) **Paris metric** ($n = 2$). If $x, y \in \mathbb{R}^2$ lie on the same ray from the origin $(0, 0)$, set

$$d_{\text{par}}(x, y) = d_2(x, y).$$

Otherwise, set

$$d_{\text{par}}(x, y) = \|x\| + \|y\|,$$

where $\|\cdot\|$ is the Euclidean norm.

(f) **Discrete metric.** Let $M \neq \emptyset$ and define

$$d(x, y) = \begin{cases} 1, & x \neq y, \\ 0, & x = y. \end{cases}$$

- (g) **The space of bounded functions with the supremum metric.** Let $M = F(a, b)$ be the set of all bounded real-valued functions on the interval $[a, b]$, where $a, b \in \mathbb{R}$ and $a \leq b$. Define

$$d_{\sup}(f, g) = \sup\{|f(x) - g(x)| : x \in [a, b]\}.$$

- (h) **Distance in an undirected connected graph.** Let $G = (V, E)$ and $M = V$. Define $d(u, v)$ to be the number of edges in a shortest path between u and v .

- (i) **Edit distance.** Let Σ be a finite alphabet of symbols, for example $\Sigma = \{a, b, c\}$. Words are finite strings of symbols; the empty word, denoted by λ , is also a word. We say that two words differ by one edit if one can be obtained from the other by:

- deleting one symbol anywhere in the word;
- inserting one symbol anywhere in the word;
- replacing one symbol by another at a given position in the word.

The distance (metric) between two words is the smallest number of edits needed to transform one word into the other.

What is a neighbourhood? What are open and closed sets? What is a ball of radius r ?

2 Unit balls

What does the unit ball, that is, the set

$$B(a, 1) = \{x \in \mathbb{R}^n : d(a, x) < 1\},$$

look like in $\mathbb{R}^n$ with each of the following metrics?

- the Euclidean metric;
- the Manhattan (sum) metric;
- the maximum metric;
- the Paris metric (in $\mathbb{R}^2$).

3 Strongly equivalent metrics

Definition: strong equivalence

Two metrics d_i, d_j on the same set M are called **strongly equivalent** if there exist $r, s \in (0, \infty)$ such that

$$r d_i(x, y) \leq d_j(x, y) \leq s d_i(x, y) \quad \text{for all } x, y \in M.$$

Show that the Euclidean, Manhattan, and maximum metrics are strongly equivalent.

Solutions

1. Examples of metric spaces

For each proposed metric, we check that its values are finite and nonnegative, that $d(x, y) = 0$ if and only if $x = y$, that it is symmetric, and that it satisfies the triangle inequality

$$d(x, y) \leq d(x, z) + d(z, y).$$

In each proof, the points x, y, z (or functions, vertices, or words) are arbitrary elements of the given space.

- (a) **Absolute value.** The number $|x - y|$ is finite and nonnegative. Also, $|x - y| = 0$ exactly when $x - y = 0$, that is, when $x = y$. Symmetry follows from $|x - y| = |-(y - x)| = |y - x|$. Finally, the triangle inequality for absolute value gives

$$|x - y| = |(x - z) + (z - y)| \leq |x - z| + |z - y|.$$

- (b) **Euclidean metric.** All distances are finite and nonnegative. A sum of squares is zero exactly when every term is zero, so $d_2(x, y) = 0$ if and only if $x_i = y_i$ for all i , equivalently $x = y$. Symmetry follows from $(x_i - y_i)^2 = (y_i - x_i)^2$. For the triangle inequality, put $u = x - z$ and $v = z - y$, so that $u + v = x - y$, and write $\|u\|_2 = (\sum_i u_i^2)^{1/2}$. By Cauchy–Schwarz,

$$\begin{aligned} \|u + v\|_2^2 &= \|u\|_2^2 + 2 \sum_{i=1}^n u_i v_i + \|v\|_2^2 \\ &\leq \|u\|_2^2 + 2 \|u\|_2 \|v\|_2 + \|v\|_2^2 = (\|u\|_2 + \|v\|_2)^2. \end{aligned}$$

Both sides are nonnegative, so taking square roots gives $\|u + v\|_2 \leq \|u\|_2 + \|v\|_2$, or $d_2(x, y) \leq d_2(x, z) + d_2(z, y)$.

Generalization to d_p , $p \geq 1$. Write $\|u\|_p = (\sum_i |u_i|^p)^{1/p}$. The sum is finite and nonnegative, and it is zero exactly when every coordinate difference is zero. Symmetry follows from $|x_i - y_i|^p = |y_i - x_i|^p$. For $p = 1$, see part (c). We use **Hölder's inequality (finite-sum form)**: for every integer $n \geq 1$, real vectors $a, b \in \mathbb{R}^n$, and real exponents $p, q \in (1, \infty)$ satisfying $1/p + 1/q = 1$, we have

$$\sum_{i=1}^n |a_i b_i| \leq \left(\sum_{i=1}^n |a_i|^p \right)^{1/p} \left(\sum_{i=1}^n |b_i|^q \right)^{1/q}.$$

For our proof with $p > 1$, choose $q = p/(p-1) > 1$; then $1/p + 1/q = 1$, as required. Put $u = x - z$, $v = z - y$, and $w = u + v = x - y$. Since $|w_i| \leq |u_i| + |v_i|$ and $|w_i|^{p-1} \geq 0$, we may multiply these coordinate inequalities by $|w_i|^{p-1}$ and sum. We then apply Hölder's inequality separately to the two sums, using $a_i = |u_i|$ and then $a_i = |v_i|$, with $b_i = |w_i|^{p-1}$ in both cases:

$$\begin{aligned}
\|w\|_p^p &= \sum_i |w_i| |w_i|^{p-1} \\
&\leq \sum_i (|u_i| + |v_i|) |w_i|^{p-1} \\
&= \sum_i |u_i| |w_i|^{p-1} + \sum_i |v_i| |w_i|^{p-1} \\
&\leq \left(\sum_i |u_i|^p \right)^{1/p} \left(\sum_i |w_i|^{(p-1)q} \right)^{1/q} \\
&\quad + \left(\sum_i |v_i|^p \right)^{1/p} \left(\sum_i |w_i|^{(p-1)q} \right)^{1/q} \\
&= (\|u\|_p + \|v\|_p) \left(\sum_i |w_i|^p \right)^{1/q} \\
&= (\|u\|_p + \|v\|_p) \|w\|_p^{p-1}.
\end{aligned}$$

Here $(p-1)q = p$, and the common factor simplifies as

$$\left(\sum_i |w_i|^p \right)^{1/q} = (\|w\|_p^p)^{1/q} = \|w\|_p^{p/q} = \|w\|_p^{p-1}.$$

If $w = 0$, the desired inequality is immediate. Otherwise, $\|w\|_p^{p-1} > 0$, so dividing by this factor gives $\|u + v\|_p \leq \|u\|_p + \|v\|_p$ (Minkowski's inequality). Returning to the original points, we obtain

$$d_p(x, y) = \|w\|_p \leq \|u\|_p + \|v\|_p = d_p(x, z) + d_p(z, y).$$

- (c) **Manhattan (sum) metric.** The sum is finite and nonnegative. Since all summands are nonnegative, it is zero if and only if $x_i = y_i$ for every i , that is, $x = y$. Each summand is unchanged when x and y are exchanged, so d_1 is symmetric. Apply the absolute-value triangle inequality in each coordinate and add the inequalities:

$$d_1(x, y) = \sum_i |x_i - y_i| \leq \sum_i (|x_i - z_i| + |z_i - y_i|) = d_1(x, z) + d_1(z, y).$$

- (d) **Maximum metric.** The maximum of finitely many nonnegative numbers is finite and non-negative. It is zero exactly when every $|x_i - y_i|$ is zero, equivalently $x = y$. Symmetry again follows from $|x_i - y_i| = |y_i - x_i|$. For every i ,

$$|x_i - y_i| \leq |x_i - z_i| + |z_i - y_i| \leq d_{\max}(x, z) + d_{\max}(z, y).$$

The right-hand side does not depend on i , so it also bounds the maximum of the left-hand side. This proves the triangle inequality.

- (e) **Paris metric.** Both formulas give finite, nonnegative values and are symmetric in x, y . Also, $d_{\text{par}}(x, x) = 0$. Notice that $d_{\text{par}}(x, y) \geq d_2(x, y)$: for points on different rays this follows from $\|x - y\|_2 \leq \|x\|_2 + \|y\|_2$. Thus $d_{\text{par}}(x, y) = 0$ implies $d_2(x, y) = 0$, hence $x = y$.

To check the triangle inequality, distinguish the following cases. If x, y lie on the same ray (including the case where one is the origin),

$$d_{\text{par}}(x, y) = d_2(x, y) \leq d_2(x, z) + d_2(z, y) \leq d_{\text{par}}(x, z) + d_{\text{par}}(z, y).$$

Otherwise, x, y are nonzero and on distinct rays. Put $r = \|x\|_2$, $s = \|y\|_2$, and $t = \|z\|_2$, so the distance to be bounded is $r + s$. If $z = 0$, the sum of the two distances is exactly $r + s$. If $z \neq 0$ is on the ray of x , then $d_{\text{par}}(x, z) = |r - t|$ and $d_{\text{par}}(z, y) = t + s$. Hence their sum is

$$|r - t| + t + s \geq r + s.$$

Here $|r - t| \geq r - t$. The case where z is on the ray of y is symmetric. If z is on neither ray, the sum is $r + 2t + s \geq r + s$. These cases exhaust all possibilities and prove the triangle inequality.

- (f) **Discrete metric.** Finiteness, nonnegativity, symmetry, and the condition for zero distance are part of the definition. If $x = y$, the triangle inequality is immediate because its left-hand side is zero. If $x \neq y$, we cannot have both $x = z$ and $z = y$. Thus at least one of $x \neq z$ or $z \neq y$ holds, so $d(x, z) + d(z, y) \geq 1 = d(x, y)$.
- (g) **Supremum metric.** Boundedness of f and g gives constants $A, B \geq 0$ with $|f(t)| \leq A$ and $|g(t)| \leq B$ for all $t \in [a, b]$. The set whose supremum is taken is nonempty and bounded above by $A + B$, since $|f(t) - g(t)| \leq |f(t)| + |g(t)| \leq A + B$. Thus the supremum exists and $0 \leq d_{\text{sup}}(f, g) \leq A + B < \infty$. Symmetry follows from $|f(t) - g(t)| = |g(t) - f(t)|$. Since $0 \leq |f(t) - g(t)| \leq d_{\text{sup}}(f, g)$ for every t ,

$$d_{\text{sup}}(f, g) = 0 \iff f(t) = g(t) \text{ for every } t \in [a, b] \iff f = g.$$

For any bounded function h and every $t \in [a, b]$,

$$|f(t) - g(t)| \leq |f(t) - h(t)| + |h(t) - g(t)| \leq d_{\text{sup}}(f, h) + d_{\text{sup}}(h, g).$$

The right-hand side is an upper bound independent of t . Taking the supremum over t therefore proves the triangle inequality. The supremum need not be attained; no continuity is assumed.

- (h) **Graph distance.** Connectivity guarantees a finite path between any two vertices. The possible path lengths are nonnegative integers, so a minimum exists. A vertex has a path of length zero to itself, while distinct vertices require at least one edge. Thus $d(u, v) = 0$ exactly when $u = v$. Because the graph is undirected, reversing a shortest path from u to v gives a path from v to u of the same length. Hence $d(v, u) \leq d(u, v)$; exchanging u, v gives the reverse inequality, and therefore symmetry. Concatenating a shortest path from u to w with one from w to v gives a walk of length $d(u, w) + d(w, v)$. Removing any cycles leaves a path no longer than this walk. A shortest path from u to v cannot be longer than this particular path, so $d(u, v) \leq d(u, w) + d(w, v)$.
- (i) **Edit distance.** A finite edit sequence always exists: delete all symbols of the first word and insert those of the second. This uses $|u| + |v|$ edits, where $|u|$ denotes the length of u . Possible numbers of edits form a nonempty set of nonnegative integers, so a finite minimum exists. Zero edits leave a word unchanged, so $d(u, v) = 0$ exactly when $u = v$. Every edit can be reversed: insertion and deletion are inverses, and a substitution can be undone. Reversing an optimal sequence gives $d(v, u) \leq d(u, v)$; exchanging the words gives equality and proves symmetry. Concatenating optimal edit sequences from u to w and from w to v gives a sequence from u to v with $d(u, w) + d(w, v)$ edits. The minimum number of edits cannot exceed this, proving $d(u, v) \leq d(u, w) + d(w, v)$.

Terminology: balls, neighbourhoods, and open and closed sets

In a metric space (M, d) , the **open ball** of radius $r > 0$ about $a \in M$ is

$$B_d(a, r) = \{x \in M : d(a, x) < r\}.$$

A **neighbourhood** of a is a set containing some open ball $B_d(a, r)$ with $r > 0$. In particular, each such ball is a neighbourhood. A set $U \subseteq M$ is **open** if every $a \in U$ has a ball $B_d(a, r) \subseteq U$ for some $r > 0$; the radius may depend on a . A set $F \subseteq M$ is **closed** if $M \setminus F$ is open. The **closed ball** is $\{x \in M : d(a, x) \leq r\}$.

2. Unit balls

In each case, substitute the formula for the metric into $d(a, x) < 1$. The inequality is strict, so points at distance exactly 1 are excluded.

- **Euclidean metric:**

$$B_{d_2}(a, 1) = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^n (x_i - a_i)^2 < 1 \right\}.$$

This is the usual open ball of radius 1 about a ; in $\mathbb{R}^2$ it is the interior of a circle, and in $\mathbb{R}^3$ the interior of a sphere.

- **Manhattan metric:**

$$B_{d_1}(a, 1) = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^n |x_i - a_i| < 1 \right\}.$$

In $\mathbb{R}^2$ this is the interior of a diamond with vertices $a \pm (1, 0)$ and $a \pm (0, 1)$: in each quadrant relative to a , the boundary equation $|x_1 - a_1| + |x_2 - a_2| = 1$ describes a straight line segment. In $\mathbb{R}^3$ it is the interior of an octahedron. In $\mathbb{R}^n$ it is the interior of the convex hull of the $2n$ points $a \pm e_i$, where e_i are the standard basis vectors.

- **Maximum metric:** A maximum is less than 1 exactly when each of its terms is less than 1.

$$B_{d_{\max}}(a, 1) = \{x \in \mathbb{R}^n : |x_i - a_i| < 1 \text{ for every } i\} = \prod_{i=1}^n (a_i - 1, a_i + 1).$$

This is an open cube with sides parallel to the coordinate axes and side length 2; in $\mathbb{R}^2$ it is the interior of a square.

- **Paris metric.** At the origin, $d_{\text{par}}(0, x) = \|x\|_2$, so $B_{d_{\text{par}}}(0, 1) = B_{d_2}(0, 1)$.

Now let $a \neq 0$, write $r = \|a\|_2$, and put $e = a/r$. For $x = te$ on the ray of a , where $t \geq 0$, the distance is $|t - r|$, so x belongs to the ball exactly when $r - 1 < t < r + 1$ and $t \geq 0$. Off that ray, the distance is $r + \|x\|_2$, so we need $\|x\|_2 < 1 - r$. Such points exist only when $r < 1$. Combining these two conditions gives

$$B_{d_{\text{par}}}(a, 1) = \begin{cases} B_{d_2}(0, 1 - r) \cup \{te : 0 \leq t < r + 1\}, & 0 < r < 1, \\ \{te : r - 1 < t < r + 1\}, & r \geq 1. \end{cases}$$

If $0 < r < 1$, the lower bound $r - 1 < t$ is automatic for $t \geq 0$. The ball consists of an open disk about the origin together with a radial segment extending through a . If $r \geq 1$, it is just an open line segment on that ray. In particular, when $r = 1$, the origin is excluded because $d_{\text{par}}(a, 0) = 1$. All these sets are open in the Paris metric, even though a line segment is not open in the Euclidean plane.

3. Strongly equivalent metrics

We seek two-sided bounds with positive constants that work for every pair of points. Let $v = x - y$ and $m = \max_i |v_i|$, so $d_{\text{max}}(x, y) = m$, $d_2(x, y) = \|v\|_2$, and $d_1(x, y) = \sum_i |v_i|$. Since at least one coordinate has absolute value m , while every coordinate has absolute value at most m ,

$$m^2 \leq \sum_i v_i^2 \leq nm^2, \quad m \leq \sum_i |v_i| \leq nm.$$

Taking square roots in the first pair of inequalities compares d_2 with d_{max} ; the second pair compares d_1 with d_{max} . To compare d_1 and d_2 directly, observe that

$$\sum_i v_i^2 \leq \left(\sum_i |v_i| \right)^2, \quad \sum_i |v_i| \leq \sqrt{\sum_i 1^2} \sqrt{\sum_i v_i^2} = \sqrt{n} \|v\|_2.$$

The first inequality follows by expanding the square: all cross terms $2|v_i||v_j|$ are nonnegative. The second is Cauchy–Schwarz applied to $(1, \dots, 1)$ and $(|v_1|, \dots, |v_n|)$. Thus, for all $x, y \in \mathbb{R}^n$,

Pairwise comparison of the three metrics

$$\begin{aligned} d_{\text{max}}(x, y) &\leq d_2(x, y) \leq \sqrt{n} d_{\text{max}}(x, y), \\ d_2(x, y) &\leq d_1(x, y) \leq \sqrt{n} d_2(x, y), \\ d_{\text{max}}(x, y) &\leq d_1(x, y) \leq n d_{\text{max}}(x, y). \end{aligned}$$

These inequalities give positive constants $r = 1$ and, respectively, $s = \sqrt{n}, \sqrt{n}, n$ in the definition of strong equivalence. The constants depend only on the fixed dimension n , not on x or y . Hence the three metrics are pairwise strongly equivalent.